BVB Campus, Vidyanagar, Hubballi – 580031, Karnataka, INDIA.

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Industry Training Report

On

C#, ASP.NET Web Forms and SQL

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering

in

COMPUTER SCIENCE AND ENGINEERING

Submitted By

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AarGees Business Solutions

Academic Year 2025–2026

BVB Campus, Vidyanagar, Hubballi – 580031, Karnataka, INDIA.

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CERTIFICATE

This is to certify that the **Industry Training Report** entitled *"Internship Training on C#, ASP.NET Web Forms and SQL"* is a bonafide record of the work carried out by **Pushpa Kalakamb (USN: 01FE22BCS205)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering** during the academic year **2025–2026**.

The student has successfully completed the internship training at **AarGees Business Solutions, Hubballi**, where she gained practical knowledge in C# programming, ASP.NET Web Forms, and SQL database management. The work presented in this report is found to be satisfactory and meets the academic standards prescribed for the course.

Internal Guide	**Head of Department**
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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **AarGees Business Solutions, Hubballi**, for providing me with the opportunity to undergo internship training and gain valuable practical exposure to industry-relevant technologies.

I extend my heartfelt thanks to the company mentors for their continuous guidance, support, and encouragement throughout the training period. Their insights and suggestions played a crucial role in enhancing my understanding of C#, ASP.NET Web Forms, and SQL.

Pushpa Kalakamb
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Abbreviations

- C# – Programming Language
- ASP.NET – Active Server Pages .NET
- SQL – Structured Query Language
- VS – Visual Studio
- SSMS – SQL Server Management Studio

Chapter 1

Introduction

1.1 Overview of Internship Training

The internship training was undertaken at AarGees Business Solutions with the objective of gaining practical exposure to modern software development technologies. The training focused primarily on C#, ASP.NET Web Forms, and SQL, which are widely used in enterprise-level application development.

Unlike classroom learning, this training emphasized understanding concepts through real-time examples and hands-on practice. The exposure to industry-relevant tools and workflows helped bridge the gap between theoretical knowledge and practical implementation.

1.2 Objectives of Training

The key objectives of the internship training were:

- To understand the fundamentals of C# programming and its application in software development
- To learn the architecture and working of ASP.NET Web Forms
- To gain hands-on experience with SQL queries and database operations
- To understand how frontend and backend systems interact in web applications
- To develop problem-solving skills using programming concepts

1.3 Organization Profile

AarGees Business Solutions is a Hubballi-based software company established in 2002. The company specializes in providing IT solutions in the education domain, along with services in e-commerce and enterprise applications.

The organization focuses on delivering cost-effective and scalable software solutions using modern technologies such as .NET and SQL Server. It has successfully served numerous educational institutions, demonstrating strong expertise in developing reliable and efficient systems.

1.4 Training Methodology

The training followed a structured approach:

- Conceptual learning through online resources (primarily YouTube)
- Practical implementation using development tools
- Regular practice with programming exercises
- Exploration of real-world use cases

This approach ensured a gradual and thorough understanding of each concept.

1.5 Scope of the Report

This report provides a comprehensive overview of the concepts and skills acquired during the internship. It covers programming fundamentals, web application development, and database management using C#, ASP.NET Web Forms, and SQL, along with their practical implementation in building dynamic applications.

Chapter 2

Tools and Technologies Used

2.1 C# Programming Language

C# is a modern, object-oriented programming language developed by Microsoft as part of the .NET framework. It is widely used for building desktop, web, and enterprise applications.

2.1.1 Key Features of C#

- Object-Oriented Programming (OOP)
- Strong type checking
- Automatic memory management
- Rich library support

2.1.2 Core Concepts Learned

Data Types and Variables

C# provides various data types such as int, float, double, char, and string. Variables are used to store data values during program execution.

Control Structures

Decision-making and looping are handled using:

- if-else statements
- switch-case
- for, while, and do-while loops

Object-Oriented Programming

The major OOP concepts include:

- Classes and Objects
- Constructors
- Encapsulation
- Inheritance

2.2 ASP.NET Web Forms

ASP.NET Web Forms is a framework used to build dynamic, data-driven web applications using a server-side programming model and event-driven approach.

2.2.1 Architecture

The Web Forms architecture follows an event-driven model where user actions (such as button clicks) trigger server-side events. It uses a page life cycle and View State mechanism to maintain state between requests, enabling seamless interaction between the user interface and server logic.

2.2.2 Key Components

- Web Forms (.aspx pages) for UI design
- Code-behind files (.aspx.cs) for application logic
- Server controls for handling user interactions
- View State for maintaining page data across requests

2.2.3 Advantages

- Supports rapid application development with built-in controls
- Simplifies database integration using ADO.NET
- Enables code reusability and separation of UI and logic

2.3 SQL (Structured Query Language)

SQL is used for managing and manipulating relational databases.

2.3.1 Types of SQL Commands

- DDL (Data Definition Language)
- DML (Data Manipulation Language)
- DCL (Data Control Language)

2.3.2 Important Concepts

- Tables and relationships
- Primary and foreign keys
- Constraints
- Joins and normalization

Command	Description
SELECT	Retrieve data
INSERT	Add new data
UPDATE	Modify existing data
DELETE	Remove data

Table 2.1: Basic SQL Commands

2.4 Development Tools

2.4.1 Visual Studio 2026

Visual Studio 2026 is used as the primary IDE for developing the application using C# and ASP.NET. It provides features like code editing, debugging, and project management.

2.4.2 SQL Server Management Studio 19

SQL Server Management Studio (SSMS) 19 is used for designing and managing the database. It supports query execution, table creation, and data handling efficiently.

2.4.3 Bootstrap 5

Bootstrap 5 is used for designing responsive and user-friendly web interfaces. It provides pre-built components and a grid system for consistent UI design.

Chapter 3

Snapshots

3.1 Development Environment

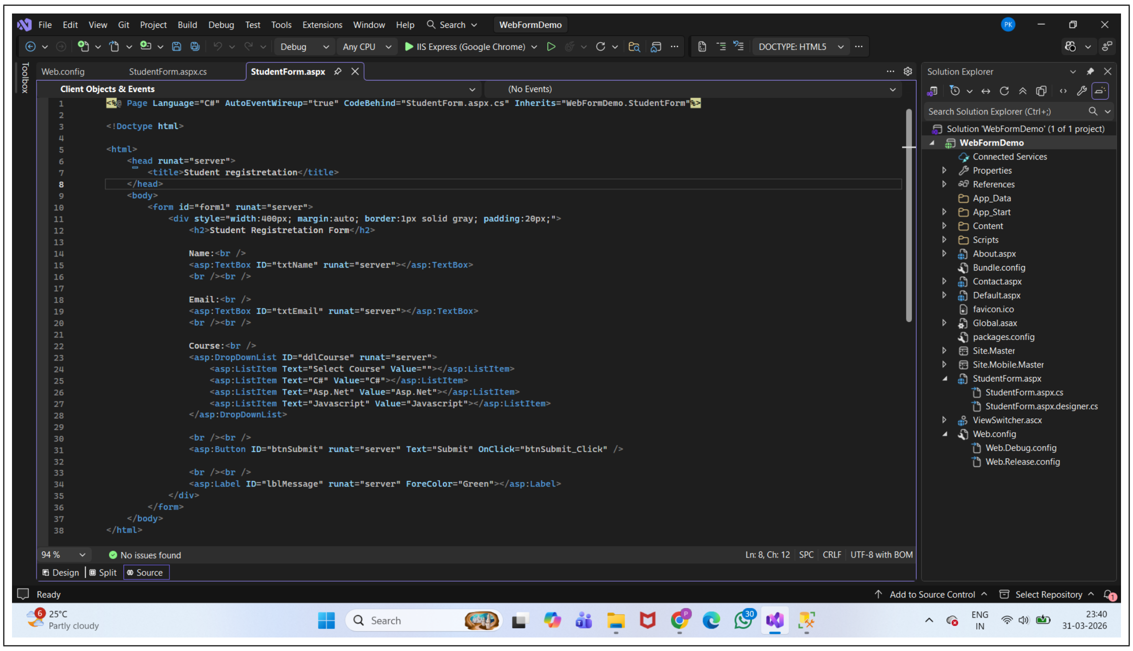

Figure 3.1: Development Environment Setup in Visual Studio

3.2 Database Management

```sql
-- Create the 'MyDatabase' database
CREATE DATABASE MyDatabase;
GO

USE MyDatabase;
GO

-- ======================================================
-- Table: customers
-- ======================================================
DROP TABLE IF EXISTS customers;
GO

CREATE TABLE customers (
    id INT NOT NULL,
    first_name  VARCHAR(50) NOT NULL,
    country     VARCHAR(50),
    score       INT,
    CONSTRAINT PK_customers PRIMARY KEY (id)
);
GO

-- Insert customers data
INSERT INTO customers (id, first_name, country, score) VALUES
    (1, 'Maria',    'Germany', 350),
    (2, ' John',    'USA',     900),
    (3, 'Georg',  'UK',      750),
    (4, 'Martin', 'Germany', 500),
```

Figure 3.2: Database Structure in SQL Server

```sql
use MyDatabase;

-- retrive all customer data
select
    country, AVG(score) as avg_score
from customers
where score != 0
group by country
having avg(score) > 430

retrive unique countries
select Distinct country
from customers

retrive only 3 customers with highest score
select top 3 *
from customers
order by score desc

-- retrive the lowest 2 customers based on score
select top 2 *
from customers
order by score asc

--get the two mosr recent orders
select top 2 *
from orders
order by order_date desc
```

Figure 3.3: Execution of SQL Queries in SQL Server

```sql
-- for not equal we can use '=!' or '<>'
select *from customers
where country != 'Germany';

select * from customers
where country <> 'Germany';

select *from customers
where score > 500;

-- logical operator
select * from customers
where country = 'USA' and score > 500;

select *from customers
where country = 'Germany' or score < 500;

select *from customers
where Not country = 'USA';


--Range operator i.e. Between it check between boundry and boundries are inclusive
--retrive customers whose score is between 100 to 500
select* from customers
where score between 100 and 500;
```

Figure 3.4: Use of SQL Operators in Queries

3.3 Web Form Interface

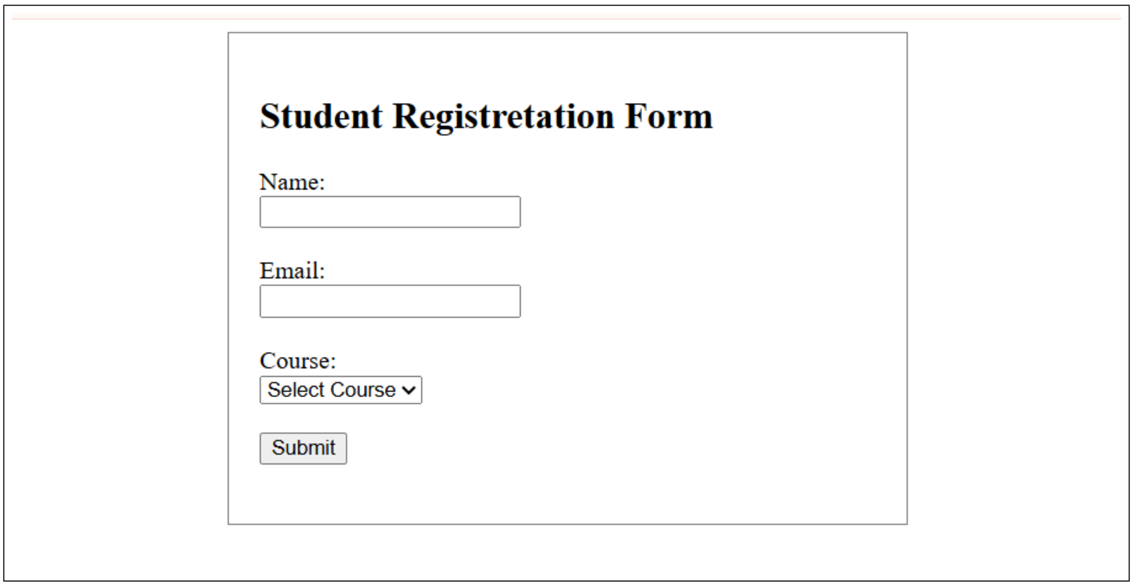

Figure 3.5: Sample ASP.NET Web Form Interface

Chapter 4

Results and Discussions

4.1 Overview of Learning

The internship training provided a solid foundation in programming and database management concepts. The hands-on approach enabled a clear understanding of how theoretical knowledge is applied to develop real-world applications and solve practical problems.

4.2 C# Programming Implementation

4.2.1 Sample Program

```
using System;

class Program
{
    static void Main()
    {
        int a = 10;
        int b = 20;
        int sum = a + b;
        Console.WriteLine("Sum = " + sum);
    }
}
```

This program demonstrates basic syntax, variable usage, and output handling in C#.

4.3 SQL Query Implementation

4.3.1 Basic Queries

```
SELECT * FROM Students;

INSERT INTO Students (Name, Age)
VALUES ('John', 21);

UPDATE Students
SET Age = 22
WHERE Name = 'John';

DELETE FROM Students
WHERE Name = 'John';
```

4.3.2 Advanced Query Example

```
SELECT s.Name, c.CourseName
FROM Students s
INNER JOIN Courses c
ON s.CourseID = c.CourseID;
```

This query demonstrates the use of joins to retrieve related data.

4.4 ASP.NET Web Form Implementation

4.4.1 Sample Web Form Code

ASP.NET Page (.aspx):

```
<asp:TextBox ID="txtName" runat="server"></asp:TextBox>
<asp:Button ID="btnSubmit" runat="server" Text="Submit" OnClick="btnSubmit_Click" />
<asp:Label ID="lblMessage" runat="server"></asp:Label>
```

Code Behind (.aspx.cs):

```
protected void btnSubmit_Click(object sender, EventArgs e)
{
    string name = txtName.Text;
    lblMessage.Text = "Hello " + name;
}
```

4.5 Discussion

The integration of C#, ASP.NET, and SQL highlights the collaborative functioning of different layers in a web application. C# is used for implementing business logic, ASP.NET manages the user interface and request handling, while SQL ensures efficient data storage and retrieval.

The training enhanced problem-solving abilities and strengthened confidence in developing and understanding real-time web applications.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

The internship training was highly valuable in enhancing both technical and practical knowledge. It provided a comprehensive understanding of C#, ASP.NET Web Forms, and SQL.

The hands-on experience gained during the training helped in developing confidence in building web-based applications.

5.2 Future Scope

- Learning ASP.NET Core for modern development
- Exploring advanced database optimization techniques
- Building full-stack web applications
- Integrating cloud technologies

Chapter 6

References

- Visual Studio Documentation: https://learn.microsoft.com/en-us/visualstudio/
- SQL Server Management Studio (SSMS) Documentation: https://learn.microsoft.com/en-us
- Bootstrap 5 Documentation: https://getbootstrap.com/docs/5.0/
- YouTube Tutorial (ASP.NET & C#): https://www.youtube.com/watch?v=AhAxLiGC7Pc

Chapter 7

Data Sheet

Username	Email	FullName	RollNumber	StreamId	LevelId	SemesterId	CourseId	SectionId	Gender	DOB	Contact
sara1	sara1@mail.com	Sara Khan	102	2	1	1	1	1	Female	15-03-2005	9876543211
bob	bob1@mail.com	Bob Smith	103	2	1	1	1	1	Male	01-01-2005	9876543210
rahul12	rahul12@mail.com	Rahul Verma	104	2	1	1	1	1	Male	12-07-2004	9876501234
priya09	priya09@mail.com	Priya Sharma	105	2	1	1	1	1	Female	22-09-2005	9876512345
aman77	aman77@mail.com	Aman Gupta	106	2	1	1	1	1	Male	05-03-2004	9876523456
neha21	neha21@mail.com	Neha Joshi	107	2	1	1	1	1	Female	18-11-2005	9876534567

Table 7.1: Sample User(Student) Data

Chapter 8

Appendices

8.1 Sample SQL Queries

The following queries were used for database operations such as retrieving and inserting data:

```
-- Select all students
SELECT * FROM Students;

-- Insert new student
INSERT INTO Students (Username, FullName, Gender)
VALUES ('rahul12', 'Rahul Verma', 'Male');
```

8.2 Database Connection (C#)

The following code snippet shows how the application connects to the SQL Server database:

```
using System.Data.SqlClient;

string connectionString = "Data Source=.;Initial Catalog=StudentDB;Integrated Securit
SqlConnection con = new SqlConnection(connectionString);
con.Open();
```

8.3 Sample C# Code

The following C# code demonstrates basic program execution:

```
using System;

class Program
{
```

```
    static void Main()
    {
        Console.WriteLine("Hello Internship");
    }
}
```

8.4 ASP.NET Web Form Snippet

The following snippet shows a simple ASP.NET Web Form control:

```
<asp:TextBox ID="txtName" runat="server"></asp:TextBox>
<asp:Button ID="btnSubmit" runat="server" Text="Submit" />
```